

Blockchains & Decentralized Finance

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Blockchain imagined by users



- distributed (without a central authority)
- secure (written information cannot be easily modified/falsified)
- information availability and transparency
- privacy preservation

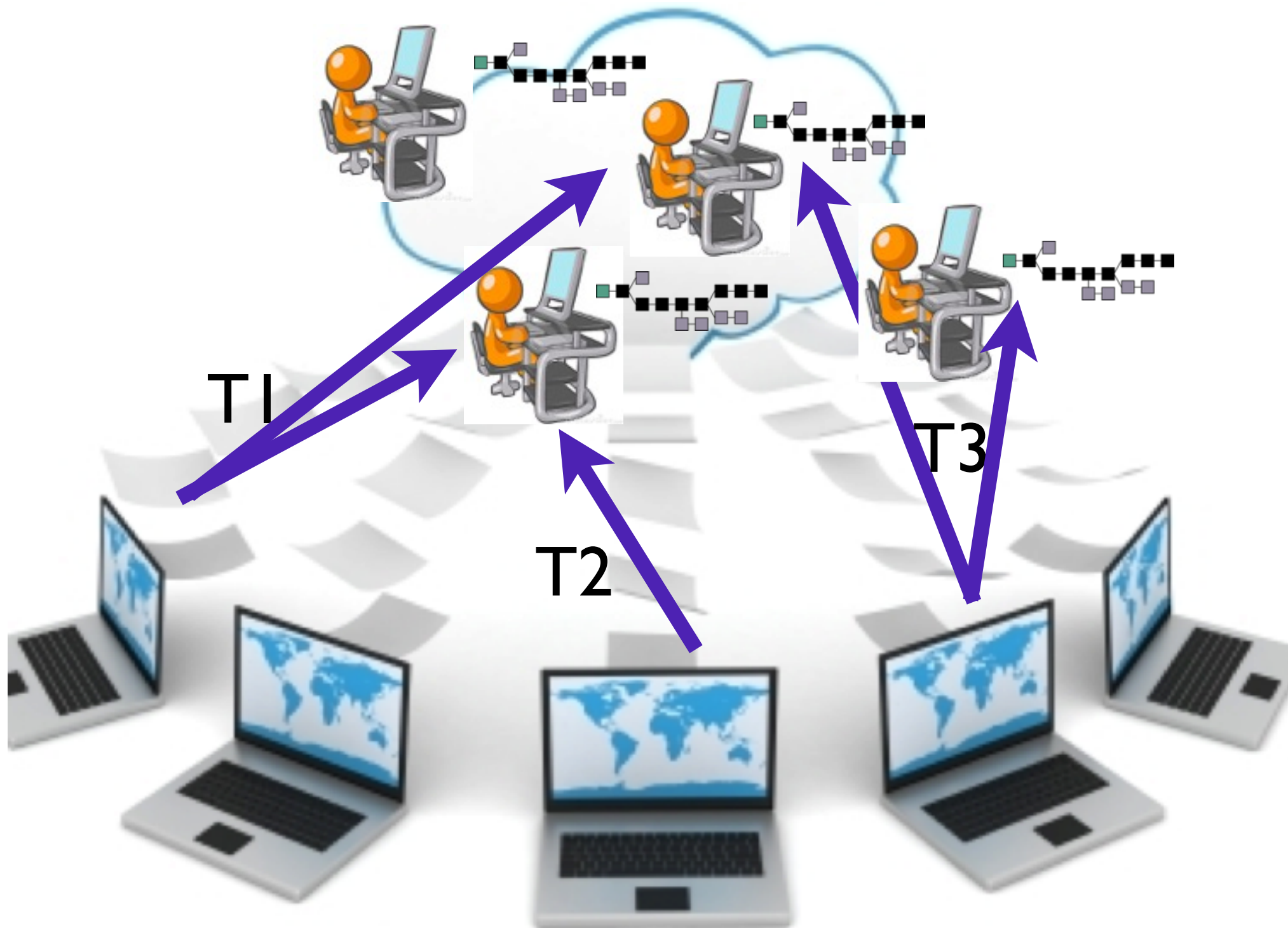
Some real blockchains



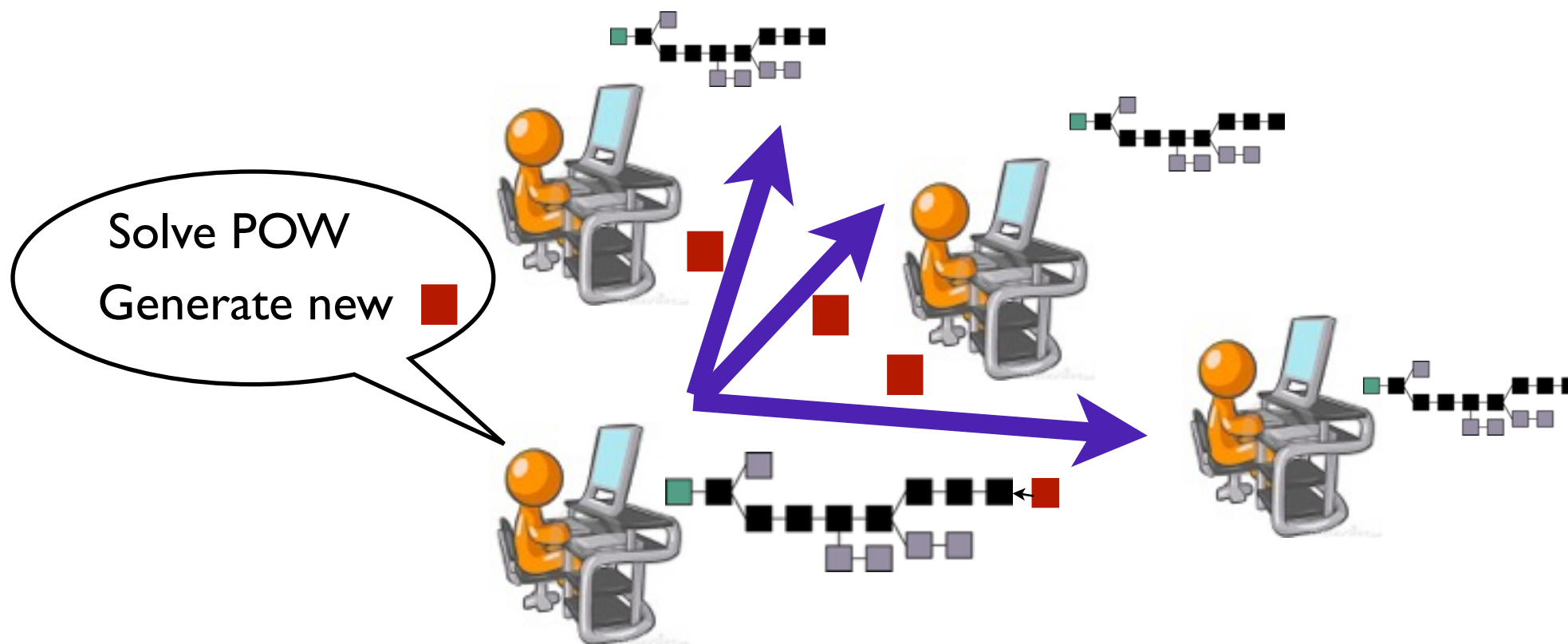
Blockchain actors

- **clients** - generate transactions
- **miners (backers)** - generate and verify **blocks** and maintain the **blockchain**
- miners and clients communicate through a **P2P network**

Bitcoin case

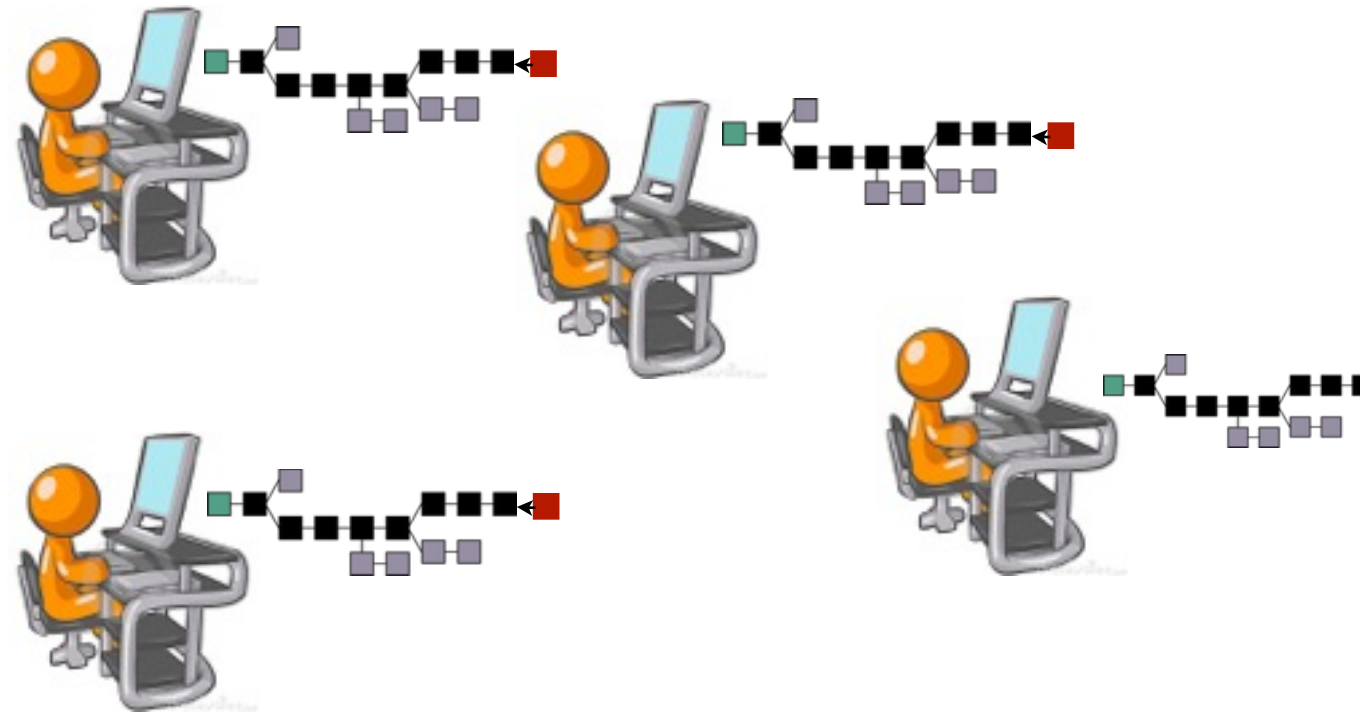


Bitcoin case



Bitcoin backers rule: mine on the longest branch.
Garanties: Agreement on a common prefix.

Bitcoin miners



Bitcoin miners rule: mine on the longest branch !

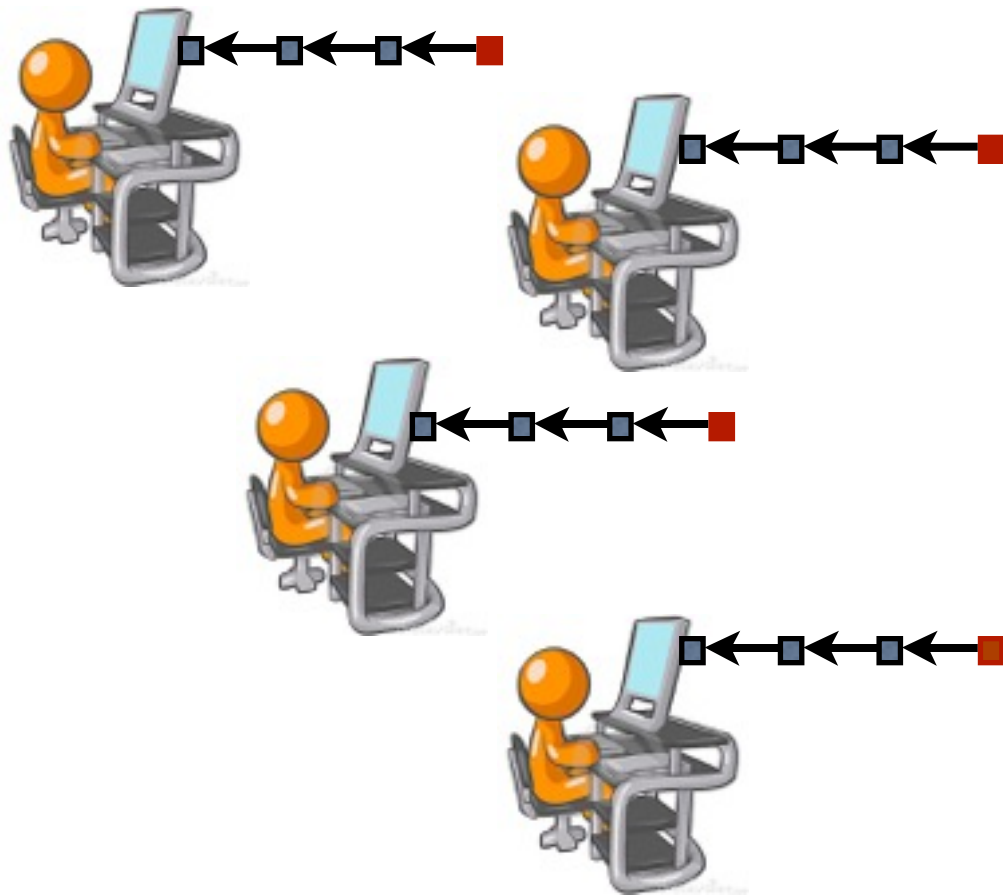
Consensus necessary for strong consistent blockchains*



* Blockchain Abstract Data Type ACM SPAA 2019

E. Anceaume, R. Ludinard, A. Del Pozzo, Maria Potop-Butucaru, Sara Tucci

Consensus-based Blockchains*



- **Agreement:** if a correct miner (baker) decides a block **b** then eventually all correct miners (backers) decide **b**
- **Validity:** A decided **b** satisfies a validity predicate
- **Integrity:** no correct miner (backer) decides twice
- **Termination:** every correct miner (backer) eventually decides

Tendermint, HyperLedger, RedBelly, Hotstuff etc

Blockchains

«agreement»

- **Non-academical**
 - POW-based: Bitcoin, Ethereum
 - DAG-based: IOTA
 - POET
- **Academical**
 - PBFT: Byzcoin, PeerCensus, Hyperledger, RedBelly, Tendermint etc
 - Probabilistic Election: Algorand

Blockchains are a Challenge !

Artificial
Intelligence

Networks

Formal
Methods

Security



Languages &
Compilation

Distributed
Systems and
Algorithms

Economy

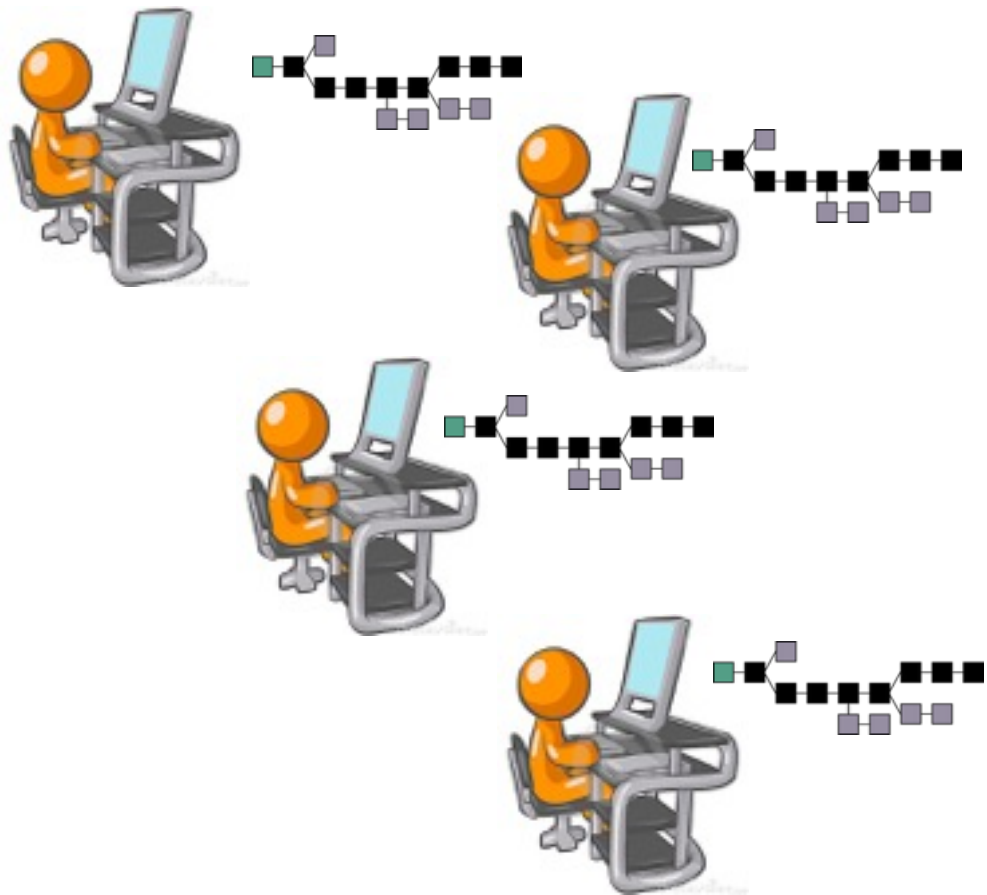
Interaction Models

Time & Causality

- Graphs
- Population Protocols
- (Markovian) Time Varying Graphs*

* Time Varying Graphs and Dynamic Networks, IJPEDES 2012
A.Casteigts, P.Flocchini, W.Quattrociocchi, N.Santoro

Behaviours* and Equilibria



- Perform verification and bake new atoms (e.g. blocks)
- Participate to the agreement (e.g. send messages)

* BAR Fault Tolerance for Cooperative Services, ACM SOSP 2015
A. Aiyer, L. Alvisi, A. Clement, M. Dahlin, J-P Martin, C. Porth

Consistency

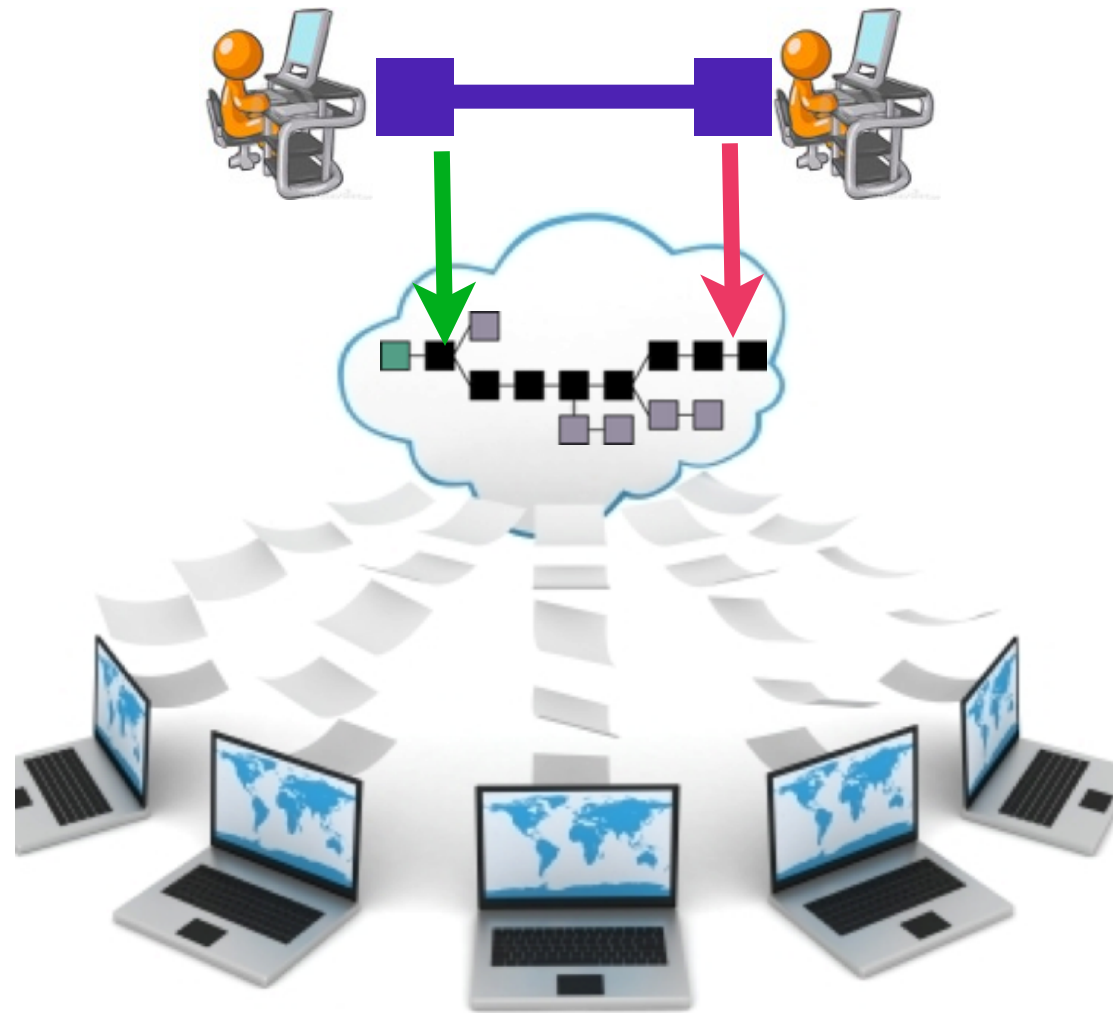


- $\text{BT-ADT} = \langle A, B, Z, \xi_0, \tau, \delta \rangle$
 - A - input alphabet : $\text{append()}, \text{read()}$
 - B - output alphabet : set of blockchains
 - Z - states : set of trees, ξ_0 - initial state : genesis block
 - $\tau : Z \times A \rightarrow Z$
 - $(\text{tree}, \text{append}(\text{block})) \rightarrow \text{tree} + \text{block}$
 - $(\text{tree}, \text{read}()) \rightarrow \text{tree}$
 - $\delta : Z \times A \rightarrow B$ - output function
 - read : returns a chain
 - $\text{append}(\text{block})$: returns «true» iff block is valid

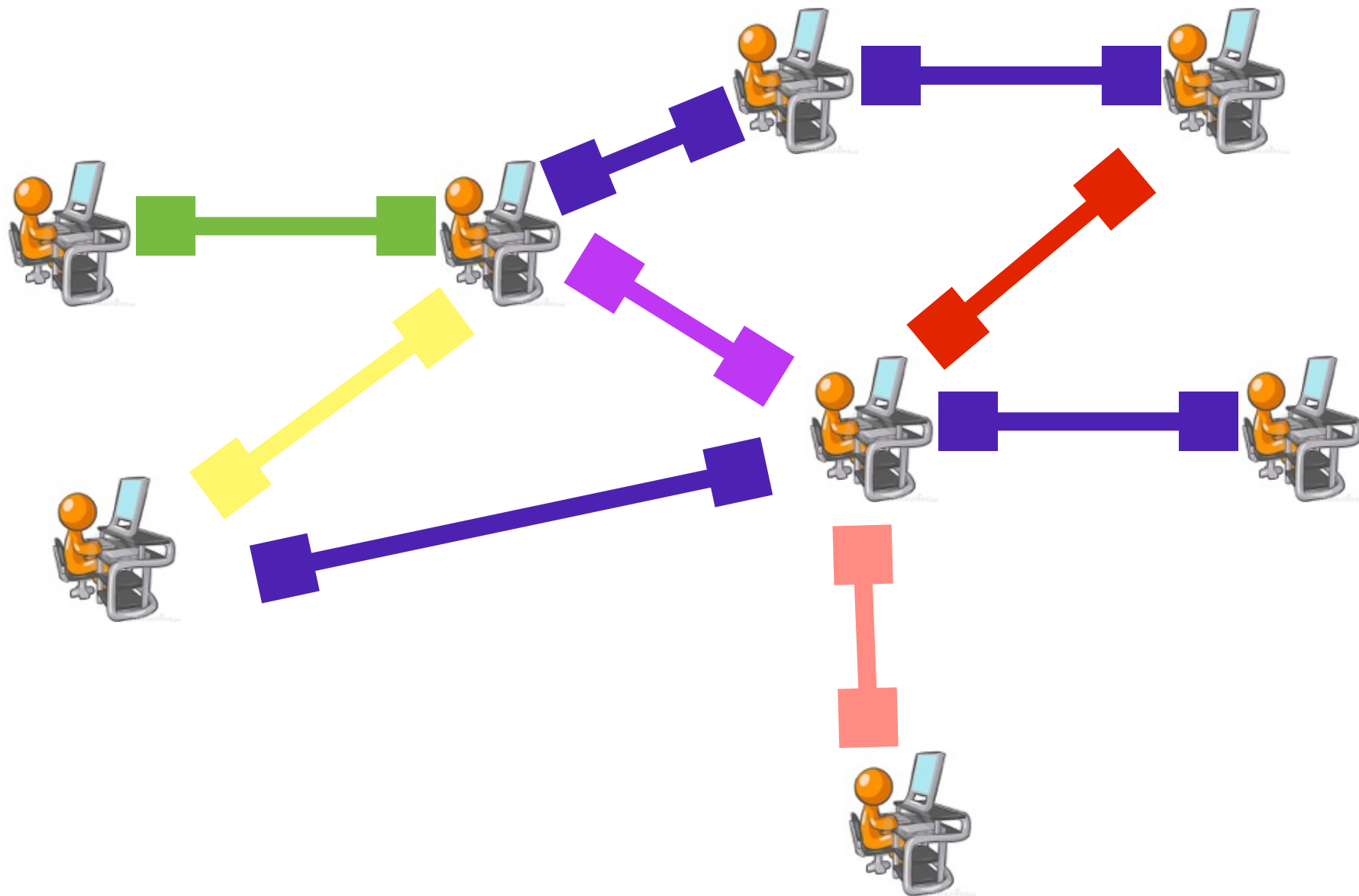
P2P Overlays



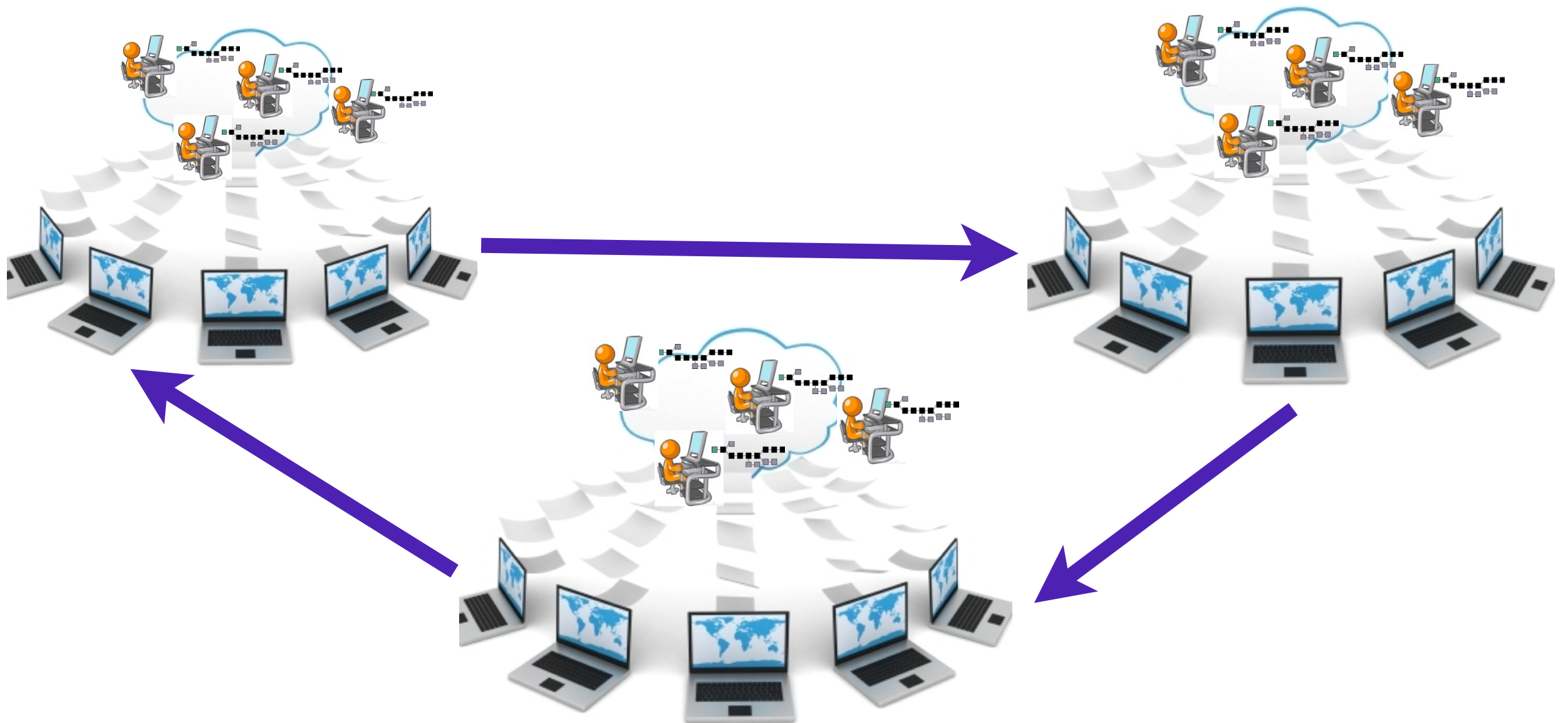
Lightning Channels



Lightning Networks



Interoperability*



* Atomic cross-chain swaps, ACM PODC 2018
Maurice Herlihy

DeFi core

- blockchains
- smart contracts
- layer 2 networking
- interoperability

